

Cloud Infrastructure for AI (AI-601)

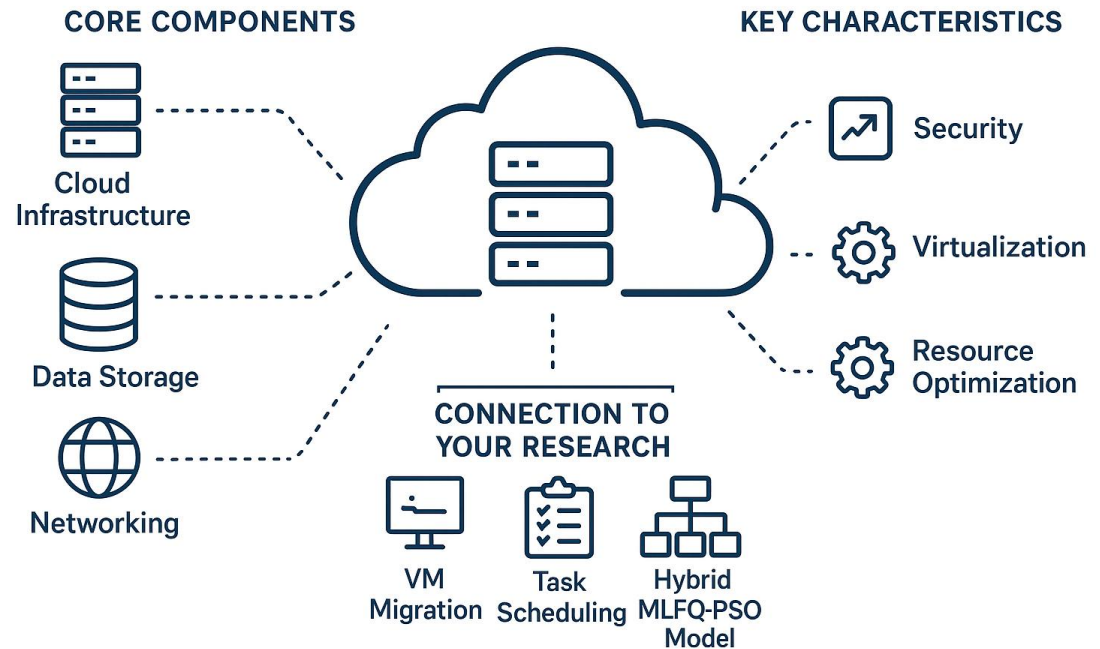
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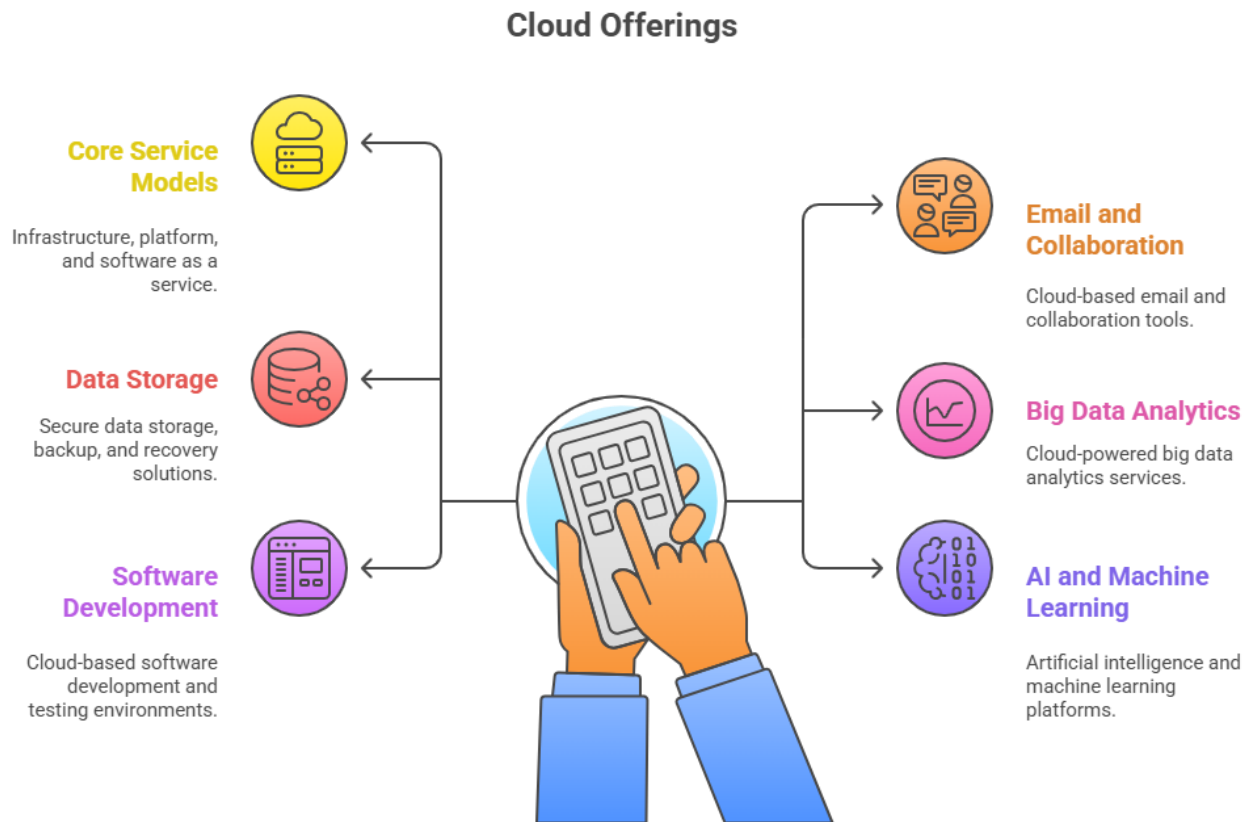
What Is Cloud Computing?

CLOUD COMPUTING

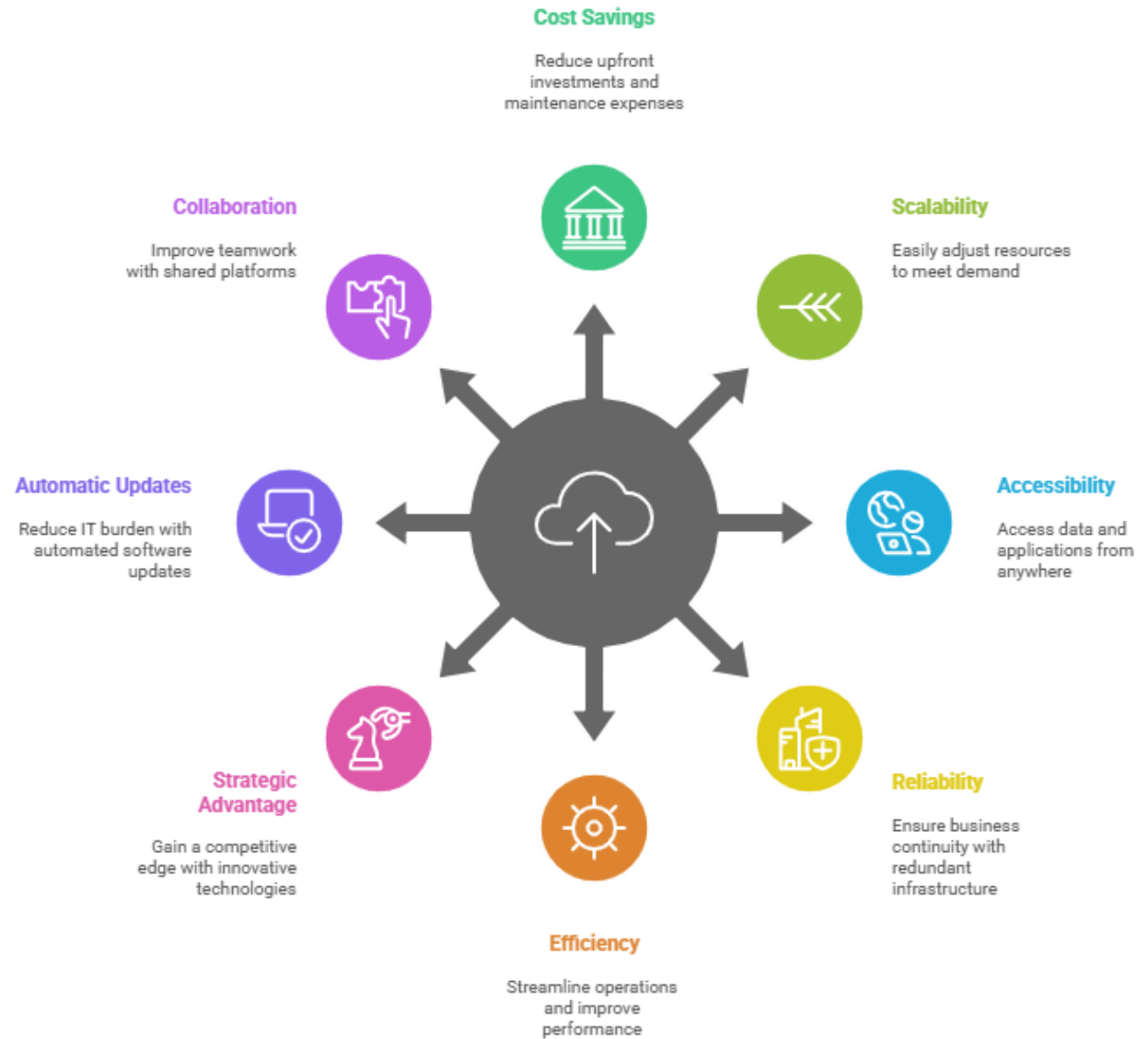


According to NIST, Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Offerings of Cloud



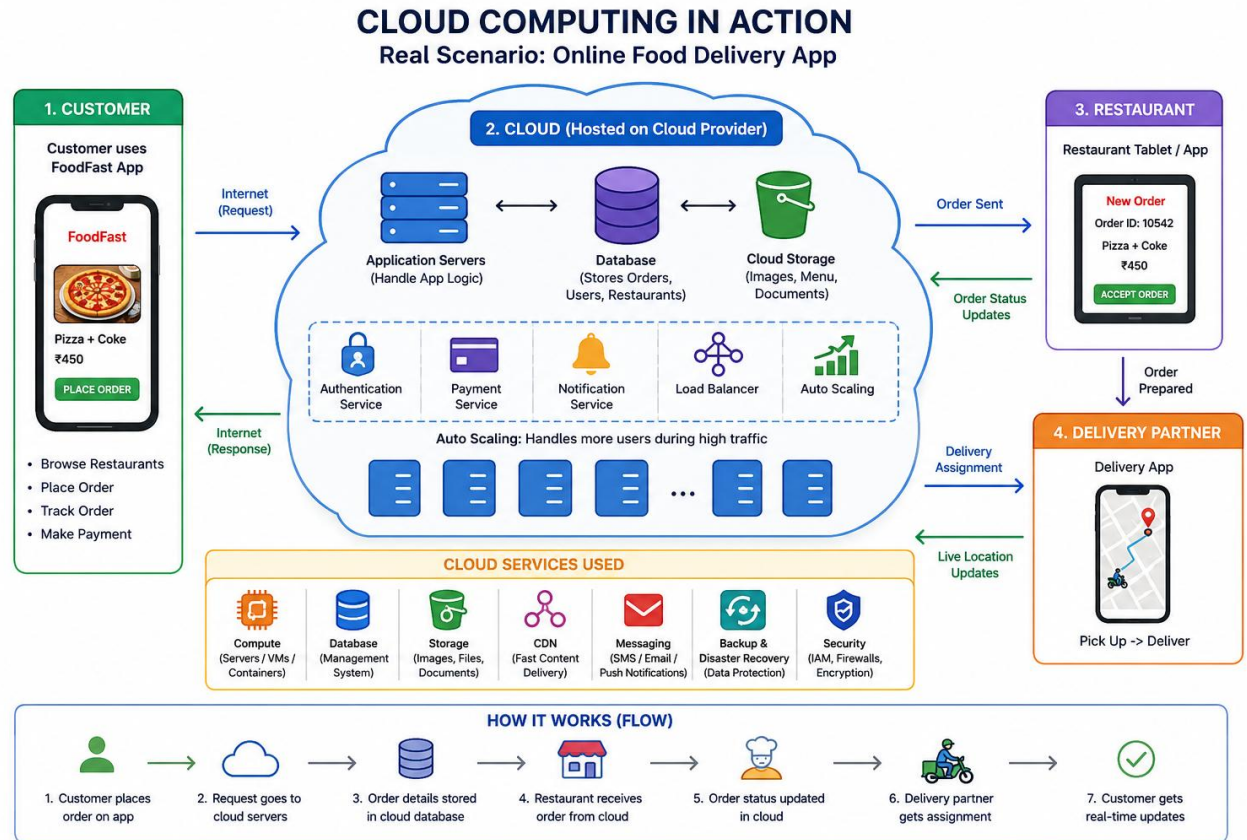
Goals of Cloud Computing



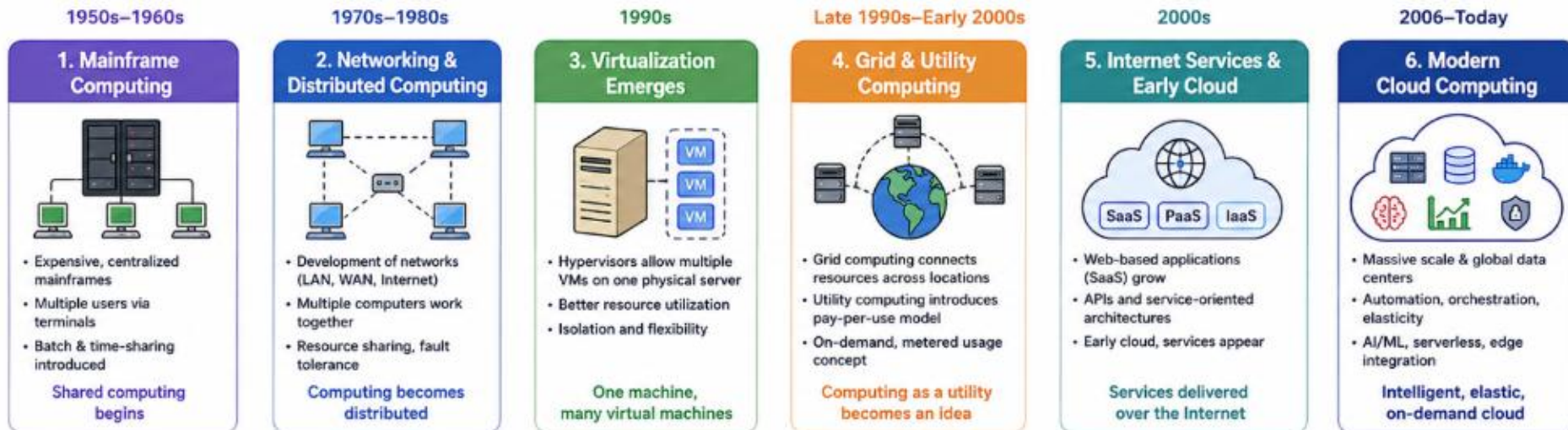
Benefits of Cloud

- Cloud computing do not need high quality equipment for user, and it is very easy to use.
- Provides dependable and secure data storage center.
- Reduce run time and response time.
- Cloud is a large resource pool that you can buy on-demand service.
- Scale of cloud can extend dynamically providing nearly infinite possibility for users to use internet.

How Cloud Computing works



Evolution of Cloud Computing



Evolution of Cloud Computing

1950s–1960s



Mainframe Computing

- Early computers were large, expensive, and shared by many users.
- The mainframe divided its processing time among multiple users.
- Shared computing resources, multiple users, centralized infrastructure, and computing as a shared resource.

1960s–1980s



Networking & Distributed Computing

- Multiple computers could cooperate instead of one doing all the work.
- Introduced resource sharing, communication between computers, distributed data, fault tolerance, and workload distribution.

1990s



Virtualization & Internet Growth

- The Internet made it possible to access computing services remotely.
- A physical server could be divided into multiple virtual machines.
- Allowed providers to share physical hardware, isolate users, increase utilization, create servers quickly, and scale easily.

1990s–2000s



Grid & Utility Computing

- Organizations combined computing resources from multiple locations for large computational problems.
- Computing viewed like a utility where users pay for what they consume.
- Introduced metered usage, on-demand resources, and service-based computing.

1999–2006



Cloud Services Emerge

- Software as a Service delivered software through the Internet instead of local installation.
- Salesforce launched its web-based CRM model in 1999.
- Amazon Web Services introduced infrastructure services like Amazon S3 and Amazon EC2 in 2006.
- Established the model of renting compute, storage, and networking over the Internet.

2008–Today



Modern Cloud Computing

- Cloud evolved from renting virtual servers to providing a huge range of managed services.
- Provides infrastructure like virtual machines, storage, and networking.
- Provides platforms like databases, containers, and application platforms.
- Provides advanced services like AI/ML, big-data analytics, serverless computing, IoT, and security.

Computing Paradigms Related to Cloud

Cloud computing did not develop in isolation. It builds upon several earlier computing paradigms.

Paradigm	Main Idea
Ubiquitous Computing	Computing is available everywhere and embedded in everyday devices
Distributed Computing	Multiple computers work together over a network
Parallel Computing	Multiple processors perform tasks simultaneously
Grid Computing	Resources from multiple organizations/sites are coordinated
Utility Computing	Computing is delivered like a utility; pay for what you use
Cloud Computing	Combines these ideas to provide scalable computing as a service

Cloud vs. Other Computing Paradigms

Feature	Cloud	Distributed	Parallel	Grid	Utility	Ubiquitous
Main focus	Services	Resource sharing	Speed	Coordination	Pay-per-use	Everywhere
Network-based	Yes	Yes	Not necessarily	Yes	Usually	Usually
Scalability	High	High	Moderate/High	High	High	High
Pay-per-use	Yes	Not necessarily	No	Usually no	Yes	No
Processing location	Cloud/DC	Multiple systems	Multiple processors	Multiple sites	Provider	Everywhere

Cloud and Other Computing Paradigms

Ubiquitous computing → Computing everywhere

Distributed computing → Many computers working together

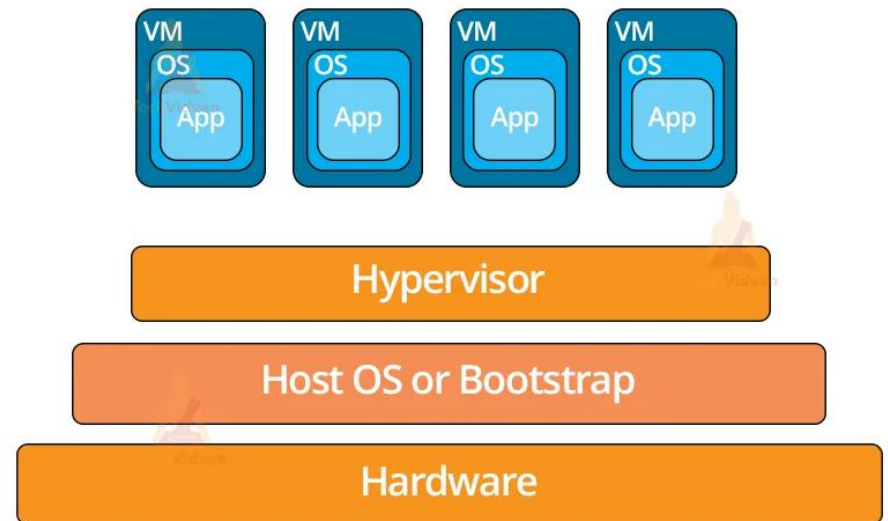
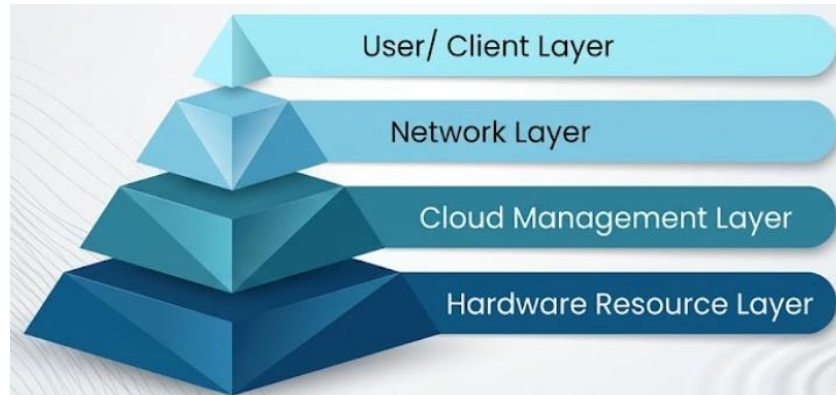
Parallel computing → Many tasks processed simultaneously

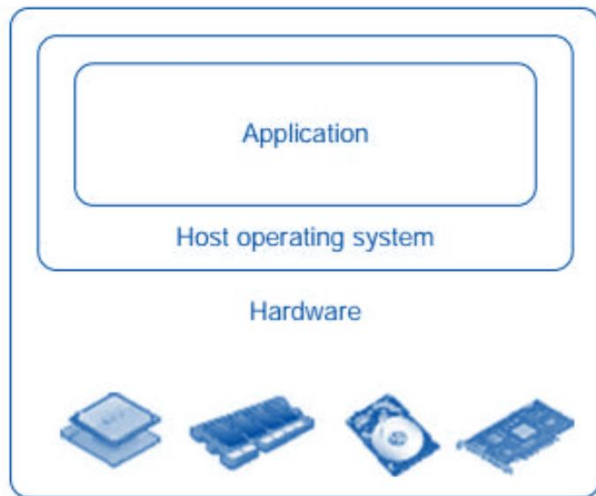
Grid computing → Coordinated resources across locations

Utility computing → Pay for what you use

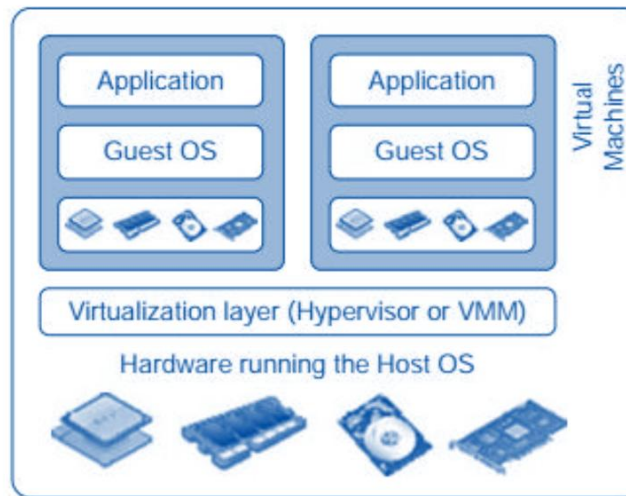
Cloud computing → Scalable computing delivered as a service

How Cloud Looks like

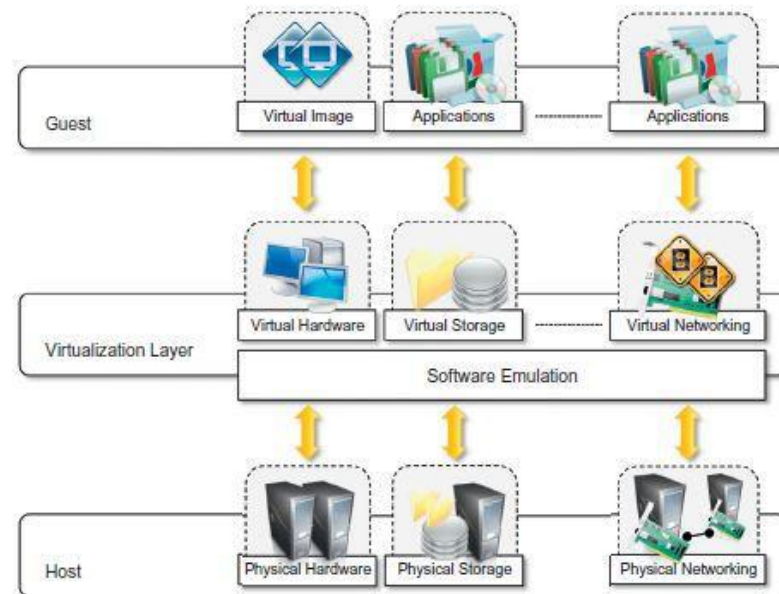




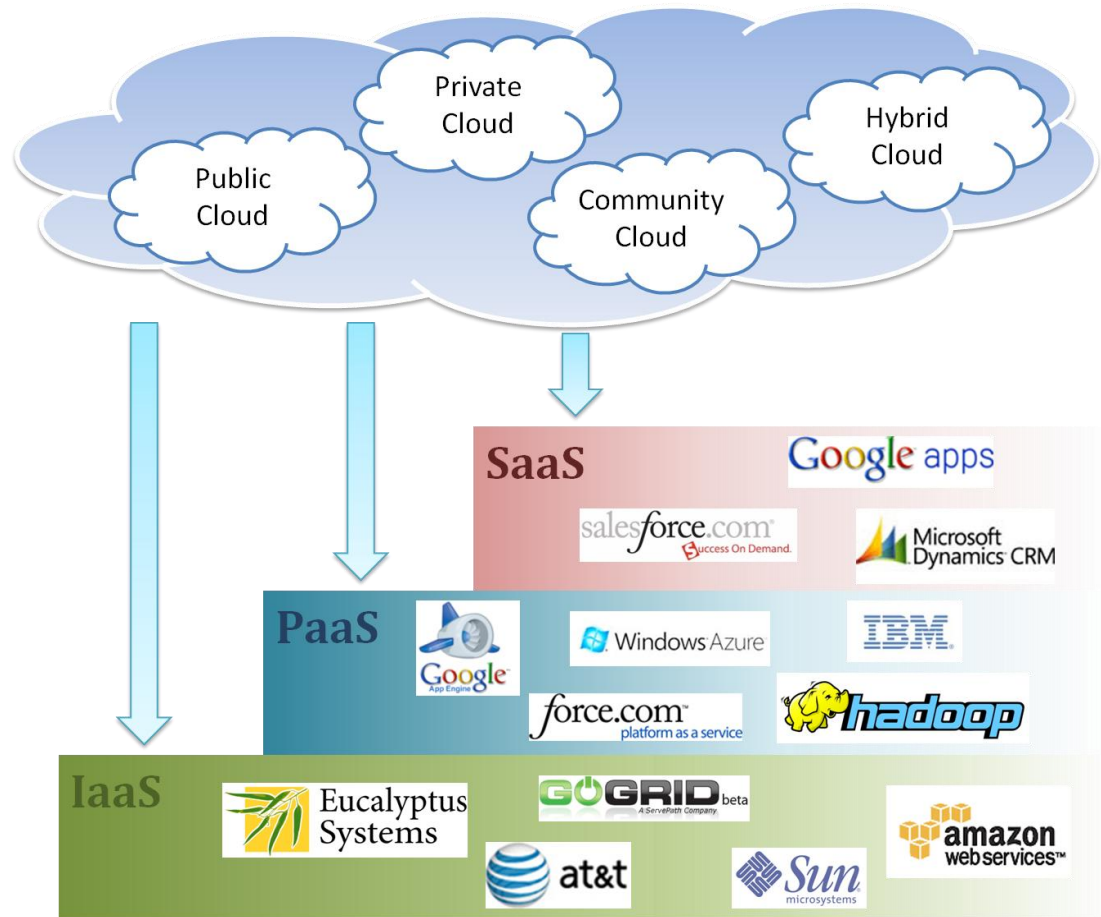
(a) Traditional computer



(b) After virtualization



Cloud Models: Service vs Deployment



◇ UNDERSTANDING CLOUD DEPLOYMENT MODELS ◇

PUBLIC CLOUD



PRIVATE CLOUD

Dedicated Enterprise Services.



COMMUNITY CLOUD

Shared Community Infrastructure.



HYBRID CLOUD

Cloud Composition.



- * Public Cloud: It is Cloud which is sold to public for storing and processing their data.
- * Private Cloud: Private Cloud is as understood is usually for granting services to enterprise which is either owned or leased.
- * Community Cloud: It is a shared infrastructure for specific community.
- * Hybrid Cloud: It is composition of two or more clouds viz. public, private or community.

CLOUD SERVICE MODELS

SOFTWARE AS A SERVICE (SaaS)

A software distribution model in which applications made available to customers over a network.



MANAGED BY
CUSTOMER
DATA ACCESS &
USER PERMISSIONS

MANAGED BY
CLOUD PROVIDER
APPLICATION, OS,
INFRASTRUCTURE

PLATFORM AS A SERVICE (PaaS)

Refers to the provision of a computing platform and deployment of associated software application solution stacks.



MANAGED BY
CUSTOMER
APPLICATIONS & DATA

MANAGED BY
CLOUD PROVIDER
RUNTIMES,
MIDDLEWARE, OS,
INFRASTRUCTURE

INFRASTRUCTURE AS A SERVICE (IaaS)

A combination of hosting, services needed to **run** cloud, **applications** made available to customers over a network.



- ✓ OPERATING SYSTEMS
- ✓ MIDDLEWARE
- ✓ RUNTIMES
- ✓ DATABASES
- ✓ DATA USAGE
- ✓ HARDWARE PROVISIONING
- ✓ COMPUTING POWER
- ✓ STORAGE
- ✓ NETWORKING

MANAGED BY
CUSTOMER
✓ OS
✓ MS

MANAGED BY
CLOUD PROVIDER
✓ NETWORKING
✓ STORAGE
✓ STOWER
✓ HARDWARE

Virtualization

Virtualization allows a **single physical computer or server** to be divided into multiple virtual resources.

Each virtual machine can:

- Run its own operating system
- Run its own applications
- Have allocated CPU and memory
- Be isolated from other virtual machines

Virtualization provides:

- Better resource utilization
- Reduced hardware costs
- Flexibility
- Faster deployment
- Easier management

Virtualization Architecture

Virtualization typically consists of three major layers:

1. Hardware or Physical Layer

Provides physical resources:

- CPU
- Memory
- Storage
- Network

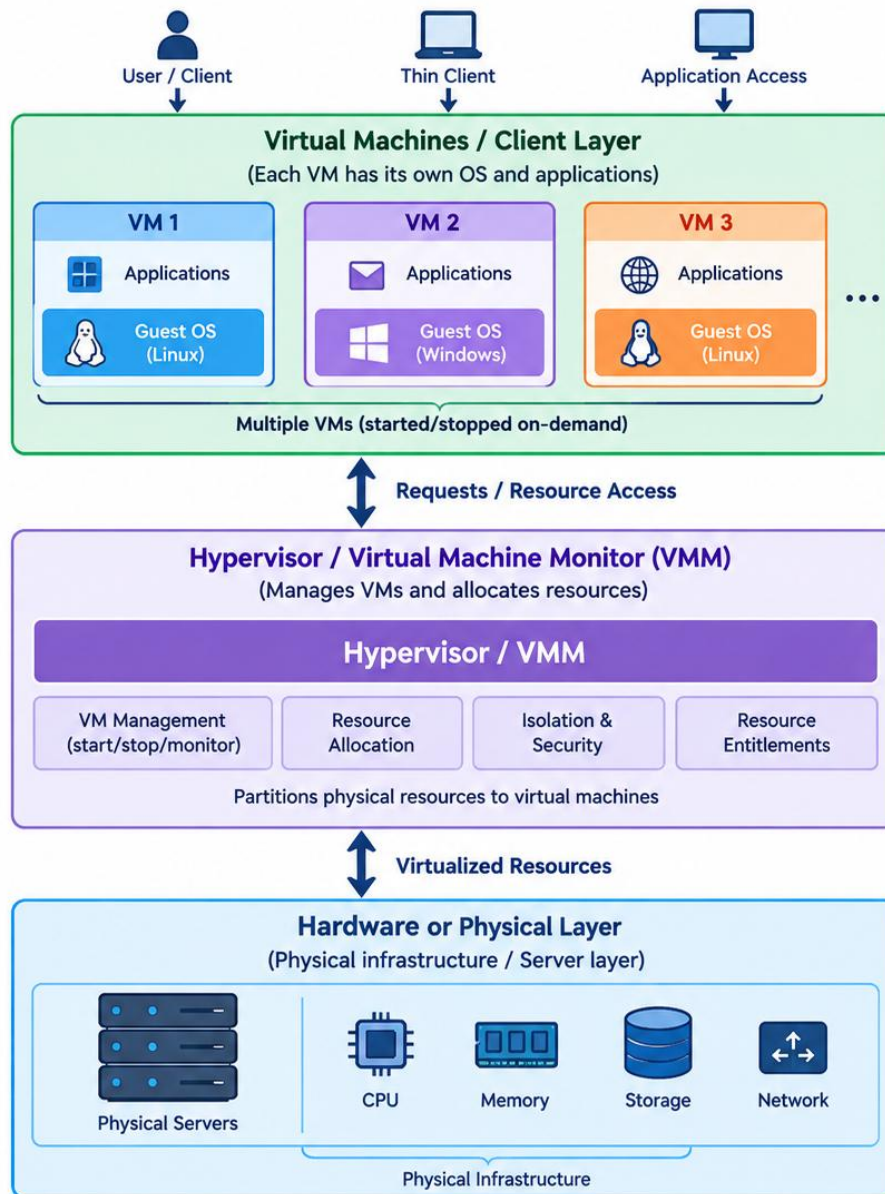
2. Virtualization Layer

The **Hypervisor or Virtual Machine Monitor (VMM)** manages and allocates physical resources.

3. Virtual Machine or Client Layer

Contains multiple virtual machines, each with:

- Guest operating system
- Applications
- Allocated virtual resources



3 Virtual Machines / Client Layer

- Operating systems that run in a virtual environment dedicated to individual users.
- Each VM contains its own OS and applications.
- Multiple VMs can be started and stopped on-demand on a single physical machine.
- Provides flexibility to configure different partitions of resources for different service requirements.

2 Hypervisor / Virtual Machine Monitor (VMM)

- Interaction interface through which clients access the hardware layer from their VMs.
- Keeps track of the availability of VMs and their resource entitlements.
- Partitions physical resources to virtual machines (each with its own OS and applications).

1 Hardware or Physical Layer

- Client interacts with the physical infrastructure for accessing resources for their applications.
- Also referred to as server layer.

Hypervisor/V irtual Machine Monitor (VMM)

- A hypervisor, or virtual machine monitor (VMM), is virtualization software that enables the creation and management of virtual machines (VMs).
- It allows to run multiple virtual machines (VMs) on a single physical machine.
- In modern computing infrastructure, hypervisors act as an intermediary between the hardware and multiple operating system (OS) instances.
- The hypervisor allocates the underlying physical computing resources, such as CPU and memory, to individual virtual machines as required.
- A hypervisor allows every virtual machine to have its own operating system and applications.

Importance of Hypervisor

- By using a hypervisor, multiple virtual machines can run on a single host computer. Each virtual machine has its own operating system and hardware resources such as a CPU, a graphics accelerator, and storage.
- They allow you to scale computing services efficiently on limited hardware infrastructure.
- Hypervisors make it possible to use more of a system's available resources and provide greater IT mobility since the guest VMs are independent of the host hardware.

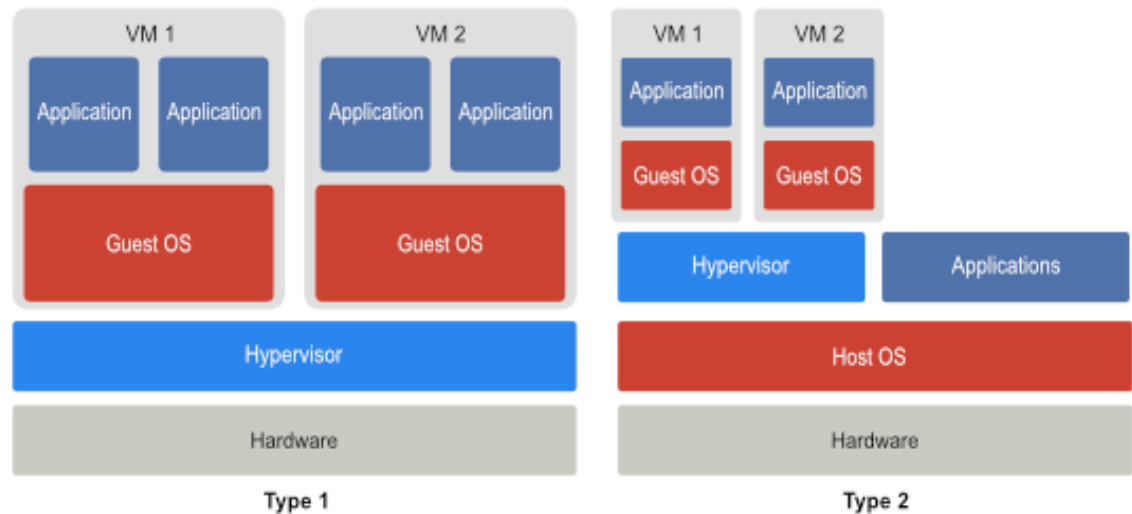
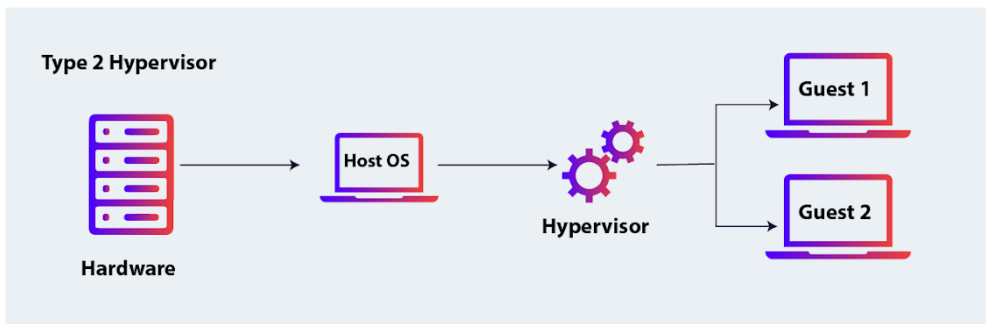
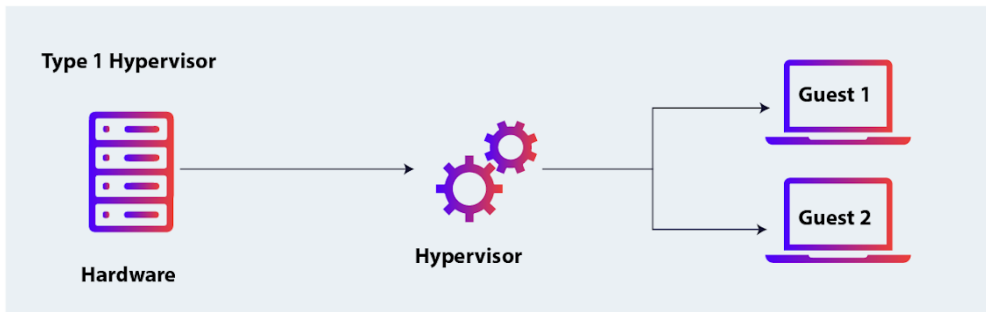
How Hypervisor works

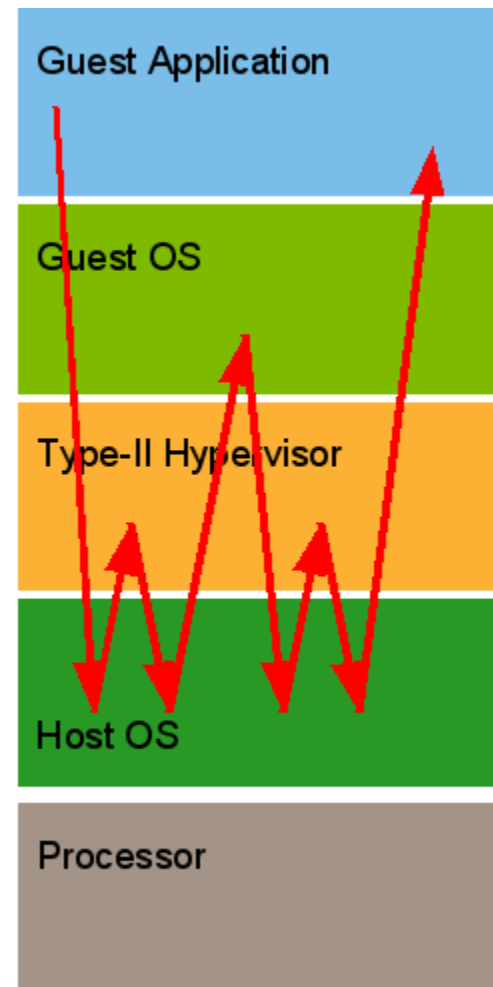
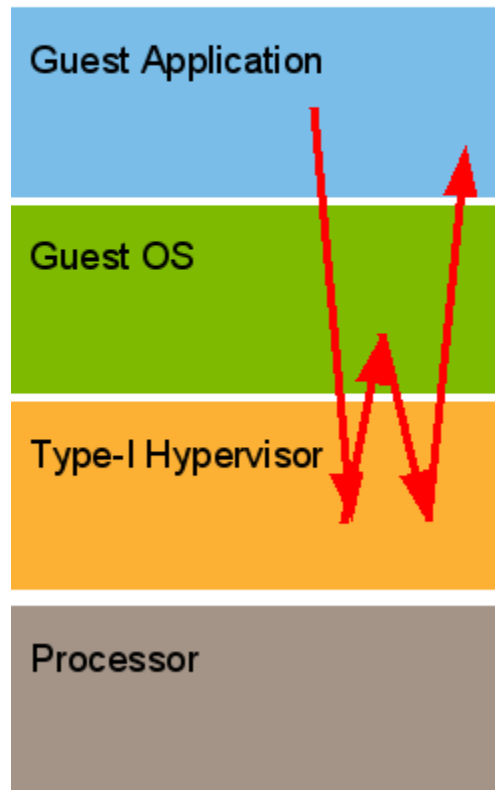
- A hypervisor provides orchestration, isolation, and resource allocation for virtual machines on a physical host.
- System administrators install the hypervisor software on a physical server.
- The hypervisor loads virtual machine images to create multiple virtual operating systems.
- It make virtualization possible by translating requests between the physical and virtual resources.
- The physical machine is known as the host, and the virtual operating systems are guest OSs.
- Applications installed on the virtual machine behave as if they're running directly on a physical computer.

Types of Hypervisor

Hypervisors can be classified into two main types, based on the type of virtualization. There are:

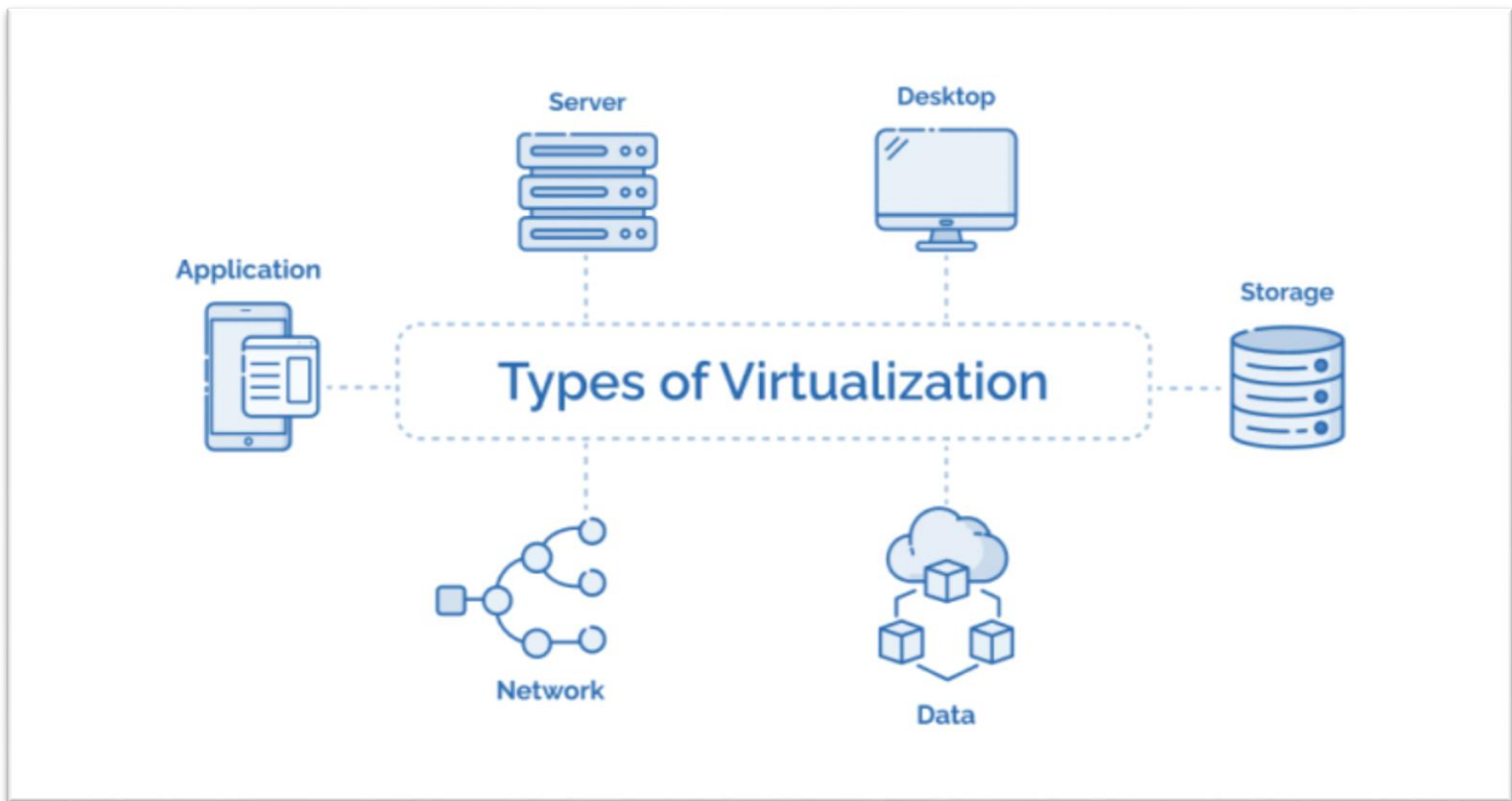
- Type 1, or “bare metal”, and
 - Type 2, or “hosted”
-
- **Type I/Bare Metal/Native Virtualization/Hypervisor:** This form of virtualization is mostly used for server virtualization in large computing systems. Type I hypervisor resides on physical host for managing the VMs. They are considered more secure and efficient as they do not require penetrating through other layers such as an operating system.
 - **Type II/Hosted/Embedded Virtualization/Hypervisor:** This form of virtualization is user or host virtualization. This type of hypervisor operates as a part of an operating system for the allocation of resources from physical machines to virtual machines.





Comparing Type-1 and Type-2

- Despite their differences, both types of hypervisors are helpful in different applications.
- Mostly, enterprise cloud data centers use type 1 or bare-metal hypervisors because of their efficiency, scalability, and flexibility when allocating resources to virtual machines. Also, a type 1 hypervisor is generally more secure and stable because it does not run on top of another operating system.
- Conversely, administrators use type 2 hypervisors because they are more user-friendly. Type 2 hypervisors are easier to install, configure, and use than bare-metal hypervisors. It is similar to installing and using other desktop applications.



Types of Virtualization

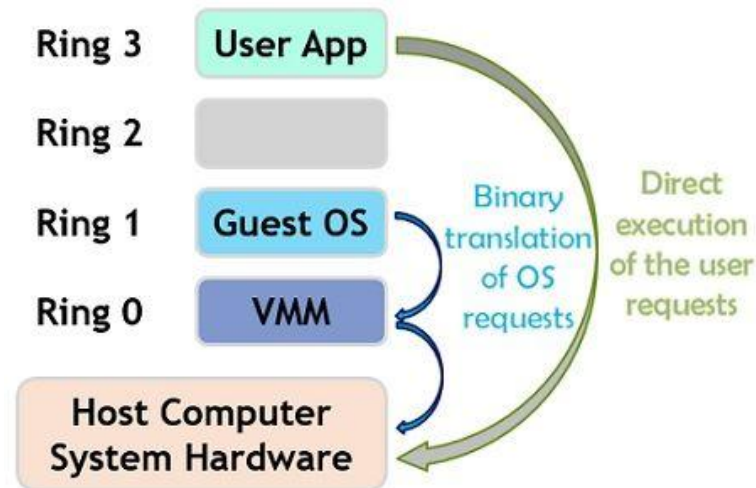
- **Hardware Virtualization:** is the broader, foundational concept. It uses a software layer called a to abstract physical computer hardware—such as the CPU, RAM, and storage—so that multiple virtual environments can run independently.
- **Server Virtualization:** Divides a single physical server into multiple independent virtual servers (Virtual Machines or VMs) to maximize hardware use.
- **Storage Virtualization:** Combines multiple physical storage devices into a single, centralized virtual storage pool for easier management.
- **Network Virtualization:** Separates network services like routing, switches, and firewalls from physical hardware to create software-defined networks.
- **Desktop Virtualization:** Hosts a user's desktop operating system on a central remote server so it can be accessed from any device.
- **Application Virtualization:** Runs applications in an isolated environment separate from the underlying operating system without requiring local installation.
- **Data Virtualization:** Creates a single unified layer to access and manage data from multiple different sources and formats.

Hardware Virtualization

- Hardware virtualization is the abstraction of computing resources from the software that uses the resources through the use of a virtual machine manager (VMM) called a hypervisor.
- The hypervisor provides the abstracted hardware to multiple guest operating systems which can then share the physical hardware resources more efficiently, increasing performance and lowering costs by enabling server consolidation.
- The main advantage of hardware virtualization is that we can create big physical servers from multiple small physical servers so that processor, memory, or other resource utilization will be effective.
- All other types of virtualizations are mainly considered under hardware virtualization.

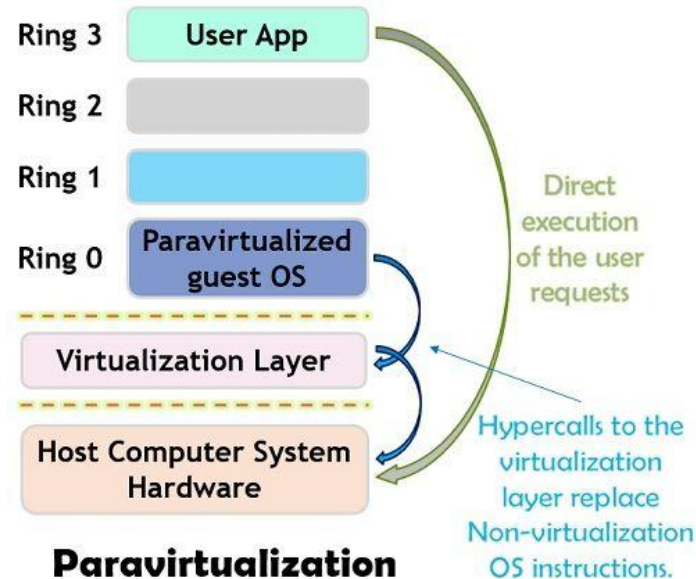
Types of Hardware Virtualization

- Major two types of hardware virtualization:
- **Full/Pure Virtualization**
 - The hypervisor completely copies or emulates the real physical hardware.
 - The guest operating system runs without any changes or knowledge that it is inside a virtual machine.
 - It uses techniques like binary translation to handle special commands, letting standard operating systems run easily.
 - It offers high compatibility and strong security isolation, but it runs slower because of the extra translation work.



• Paravirtualization

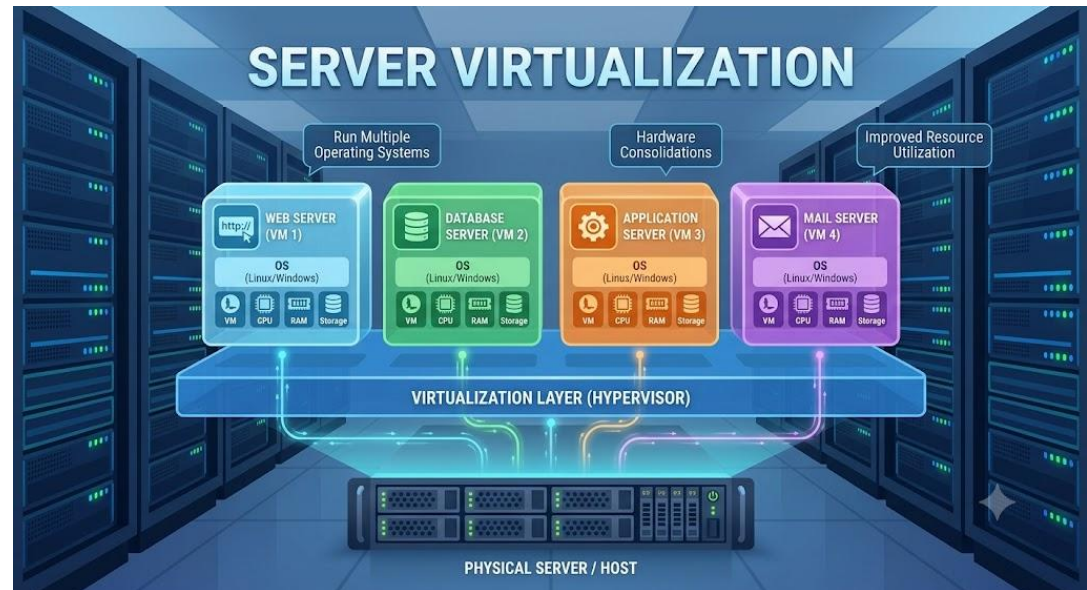
- The hypervisor provides a streamlined system interface rather than a full hardware copy
- The guest operating system's kernel is changed so it knows it is virtualized.
- Instead of sending regular hardware requests, the modified operating system uses special calls—called **hypercalls**—to talk directly to the hypervisor.
- It is faster and performs better than full virtualization, but it works with fewer operating systems and is less portable



FULL VIRTUALIZATION	PARAVIRTUALIZATION
A common and cost effective type of virtualization in which computer service requests are separated from the physical hardware that facilitates them	An enhancement of virtualization technology in which a guest OS is recompiled prior to installation inside a virtual machine
Allows the guest operating systems to execute independently	Allows guest operating systems to communicate with the hypervisor
Guest operating system issues hardware calls to access hardware	Guest operating system directly communicate with the hypervisor using drivers
Lower Performance	Higher Performance
	Visit www.PEDIAA.com

Server Virtualization

- This specifically focuses on partitioning a single physical server into multiple isolated virtual servers (Virtual Machines or VMs).
- It applies hardware virtualization principles strictly to server infrastructure to maximize computing resources.
- server virtualization is just hardware virtualization focused on server hardware rather than client computers or other equipment.



Network Virtualization

- In network virtualization, network resources like firewalls, load balancers, network interface cards, bandwidth, and so on are divided logically into multiples chunks. And these multiple chunks work independently from each other, and in real-time, these chunks were assigned to any device.
- Network virtualization creates its virtual network in which each application sees its own logical network and independent from the physical network. Virtual LAN was an example of Network virtualization.

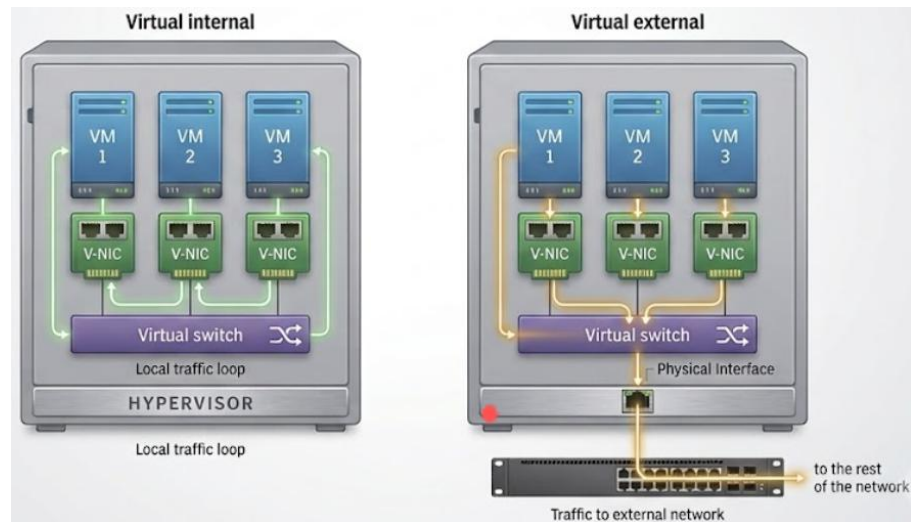
Types of Network Virtualization

- **Internal network virtualization**

If services were provided to the users when a single system connected with different resources within the systems, it would be referred to as internal network virtualization.

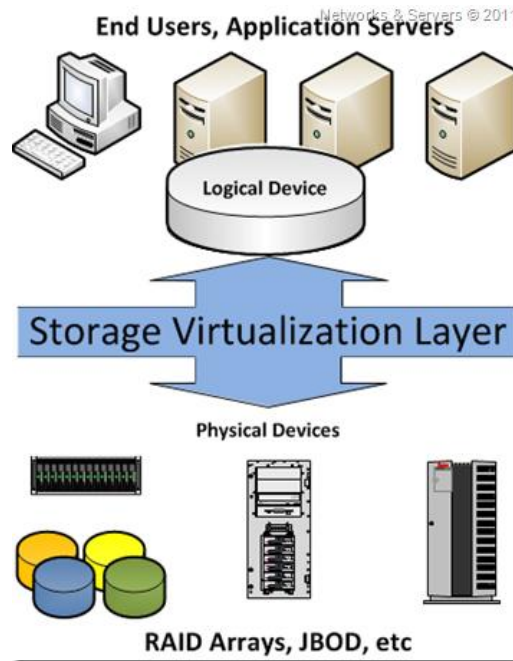
- **External network virtualization**

When multiple systems are connected with multiple networks, and users were accessing the resources from there, it was called external network virtualization.



Storage Virtualization

- Storage virtualization combines multiple physical hard drives or storage arrays into a single, centralized pool that you can manage as one large unit.
- In this type of virtualization, data were pooled into main servers from various storage devices that appeared as a single device. And all the control of the storage device was in the hand of that main server.

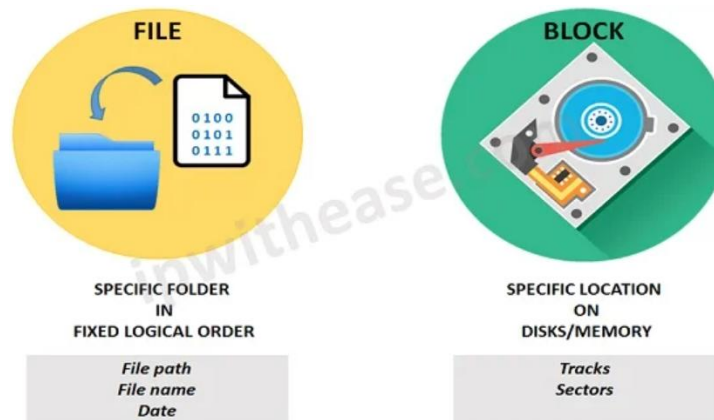


Storage virtualization uses:

- a **separation layer** that hides the details of physical storage from applications and operating systems.
- Physical disks are divided into small blocks and combined into a **shared storage pool**.
- A **metadata mapping** mechanism keeps track of where each data block is physically stored, allowing applications to access data without knowing its actual location.

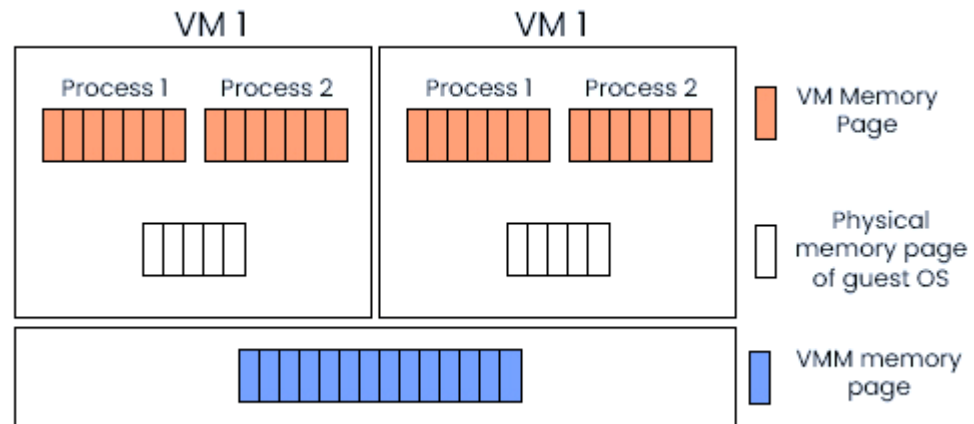
Types of Storage Virtualization

- **Block-level virtualization** – Combines physical storage blocks from multiple disks into a virtual storage pool. The operating system accesses the storage as a single logical device.
- **File-level virtualization** – Provides a unified view of files stored across different file servers or storage systems, allowing users and applications to access them through a common namespace.



Memory Virtualization

- Memory virtualization decouples physical RAM from operating systems or applications, creating an abstracted pool of memory resources.
- In memory virtualization, memory areas of different servers available on the cloud were consolidated to form a single memory pool. From that consolidation memory areas, users access or use their services.



Types of Memory Virtualization

1. Virtual Memory

- The operating system uses **virtual addresses** instead of directly accessing physical RAM.
- Provides each VM/process with its own virtual address space
- Maps virtual addresses to physical memory
- Provides isolation between VMs

2. Memory Overcommitment

- The hypervisor allocates more virtual memory to VMs than the available physical RAM.
- Useful when VMs do not use all their allocated memory simultaneously
- Improves physical memory utilization
- May use swapping or other techniques when memory demand increases

3. Memory Ballooning

- A **balloon driver** inside the VM allows the hypervisor to reclaim unused memory from one VM and allocate it to another VM.
- Dynamically adjusts memory allocation
- Improves resource utilization
- Common in virtualized cloud environments

Types of Memory Virtualization

4. Page Sharing / Memory Deduplication

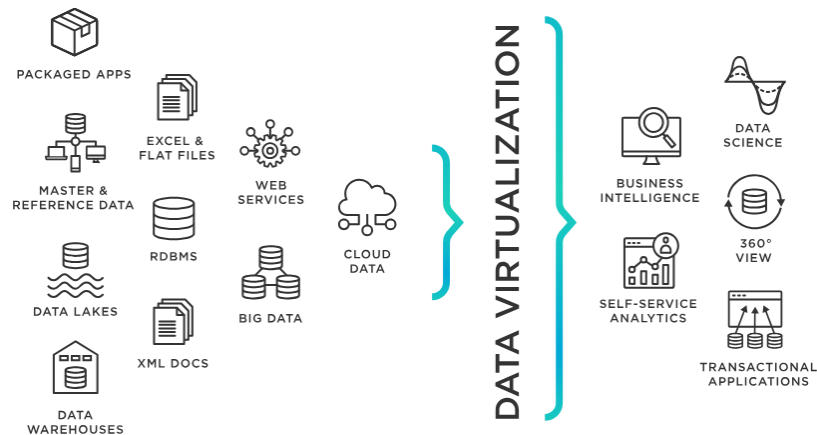
- Identical memory pages used by different VMs are stored **only once** in physical memory.
- Reduces duplicate memory usage
- Increases the number of VMs that can run on a server
- Particularly useful when VMs run similar operating systems or applications

5. Memory Swapping

- When physical memory is insufficient, less-used memory pages can be moved from RAM to **disk/storage**.
- Frees physical RAM
- Allows VMs to continue operating during memory pressure
- Slower than accessing RAM

Data Virtualization

- Data virtualization is a technology that creates an abstract layer to let users access and query data from multiple diverse sources as if it were in a single place.
- It provides a single, unified view of data from different sources without physically moving or copying all the data into one location.
- Suppose you want information for some field, and instead of searching that information in different areas, you wish to explore all this at one place, and this happens or is done with the help of data virtualization. It collects all the data from multiple sources and consolidates that multiple information into a single form so that users can get it at one place.



Types of Data Virtualization

1. Database Virtualization

- Combines or provides access to data from multiple databases through a **virtual layer**.
- Hides differences between databases
- Provides a unified view
- Useful for integrating distributed databases

2. File/Data Virtualization

- Provides a **single logical view** of files or data stored across different systems.
- Users access data without knowing its physical location
- Simplifies access to distributed data

Types of Data Virtualization

3. Data Warehouse Virtualization

- Provides virtual access to data from multiple sources for **reporting and analytics** without necessarily copying all data into a traditional warehouse.

4. Data Lake Virtualization

- Provides a unified view of structured, semi-structured, and unstructured data stored across different data lakes and storage systems.

5. Cloud Data Virtualization

- Integrates data across **multiple cloud platforms, on-premises systems, and applications** through a common access layer.

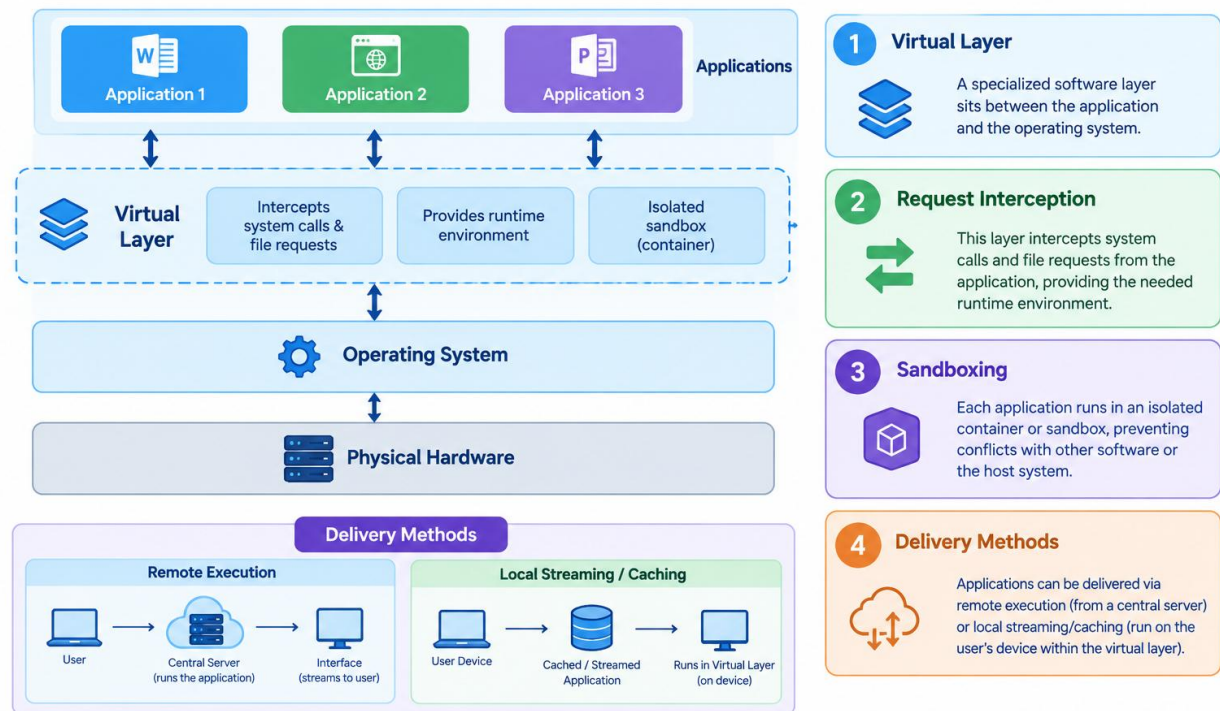
Desktop Virtualization

- **Desktop virtualization** is a technology that separates a user's **desktop environment**—operating system, applications, and settings—from the physical device. The desktop runs on a **central server or cloud infrastructure** and can be accessed remotely from a laptop, desktop, tablet, or other device.
- It separates a user's operating system and desktop environment from the physical device they use to access it.
- In the case of desktop virtualization, you can easily access your device in which you are working from any third place or from any distant location. This type of virtualization makes the clones of your complete system, which was put into a particular cloud server location with the use of various tools such as citrus, Xen Desktop.



Application Virtualization

- Application virtualization is a software technology that separates a computer program from the underlying operating system so it can run without a traditional local installation.



Virtualization Type	Physical Resource Abstracted	Main Benefit
Server	CPU & RAM	Maximizes computer hardware use
Network	Switches, Routers & Bandwidth	Automated, flexible network setup
Storage	Hard Drives & Flash Arrays	Unified, easy-to-manage storage pools
Desktop	User Workstations	Secure, centralized remote access
Application	OS Environment	Prevents app conflicts, runs anywhere
I/O	Hardware Ports & PCIe	High-speed data paths for VMs

Benefits of Virtualization

- **Cost Efficiency** – Reduces the need for multiple physical servers, lowering hardware and maintenance costs.
- **Scalability** – Easily scale resources up or down based on demand.
- **Flexibility** – Run multiple operating systems and applications on the same hardware.
- **High Availability** – Ensures business continuity with backup and failover support.
- **Resource Optimization** – Maximizes the use of available hardware resources.
- **Improved Security** – Virtual environments can be isolated, reducing the risk of cyber threats spreading.

Service Level Agreement (SLA)

- A **Service Level Agreement (SLA)** is a formal agreement between a **cloud service provider** and a **customer** that defines the services to be provided and the expected level of service.
- It specifies:
 - **Availability/Uptime**: How much time the service should remain available, e.g., 99.9%.
 - **Performance**: Expected response time, throughput, or processing capacity.
 - **Security**: Security measures and responsibilities of the provider.
 - **Data Protection**: Backup, recovery, and data-handling requirements.
 - **Support**: Technical support availability and response times.
 - **Disaster Recovery**: Procedures for recovering services after failures.
 - **Penalties/Service Credits**: Compensation if the provider fails to meet agreed targets.
- SLA provides **clear expectations and accountability** between the customer and cloud provider.

Types of SLAs

- **1. Customer-based SLA**
Agreement designed for a specific customer and their requirements.
- **2. Service-based SLA**
A standard SLA applied to all customers using a particular service.
- **3. Multi-level SLA**
Different SLA levels are defined for different customers, departments, or services.



Components of SLA

- **Agreement Overview** – An agreement overview is an element that includes individuals who are involved, service effective, and expiry date. Also, it contains a general statement about the details of the agreement.
- **Goals and Objectives** – It explains the goals and objectives, and outlines the ability to obtain a mutual agreement.
- **Stakeholders** – The stakeholders hold the name of the parties involved in the agreement—for example, a tech support company and a game-developing firm.
- **Periodic Review** – Along with the goals and objectives, it defines the parameters regarding the review timelines of a particular SLA.
- **Responsibilities of Each Party** – This section outlines and declares the responsibilities of each party.
- **Penalties for Breach** – The indemnification parameters for any unwanted data or property breach (penalties/compensations) will be mentioned in this area.
- **Protocol for Adding/Updating the Agreement** – It describes all the protocols for adding or updating the agreement.

Benefits of SLA

- It ensures enhanced customer satisfaction.
- Helps solidify the trust.
- Projects an established source of information.
- Boosts the overall working experience with a clearly defined goal.
- Brings forth a better experience for employees as well.

Cloud Security and Compliance

Cloud Security:

- Protecting cloud-based **data, applications, infrastructure, and services** from unauthorized access, attacks, and failures.
- It refers to the technologies, policies, controls, and practices used to protect:
 - Cloud data
 - Applications and services
 - Virtual machines and containers
 - Networks and infrastructure
 - User identities and access

Cloud Compliance:

- Ensuring that cloud services and data handling follow **laws, regulations, standards, and organizational policies**.
- It ensuring that cloud systems and data follow:
 - Government laws and regulations
 - Industry standards
 - Organizational policies
 - Data protection and privacy requirements

- **Main goals include:**

- *Confidentiality* – only authorized users can access data.
- *Integrity* – data remains accurate and unmodified.
- *Availability* – services and data remain accessible when required.
- *Privacy* – personal and sensitive information is properly protected.
- *Accountability* – activities can be monitored and traced.

Cloud Security Risks

- Data breaches
- Unauthorized access
- Account hijacking
- Malware and ransomware
- Insecure APIs
- Misconfigured cloud resources
- Insider threats
- Data loss
- Denial-of-service attacks
- Weak passwords and stolen credentials

Cloud resources can be accessed through the internet from many locations and devices. Therefore, organizations must carefully control who can access resources, what they can access, and how they access them.

Security Monitoring and Auditing

Monitoring activities

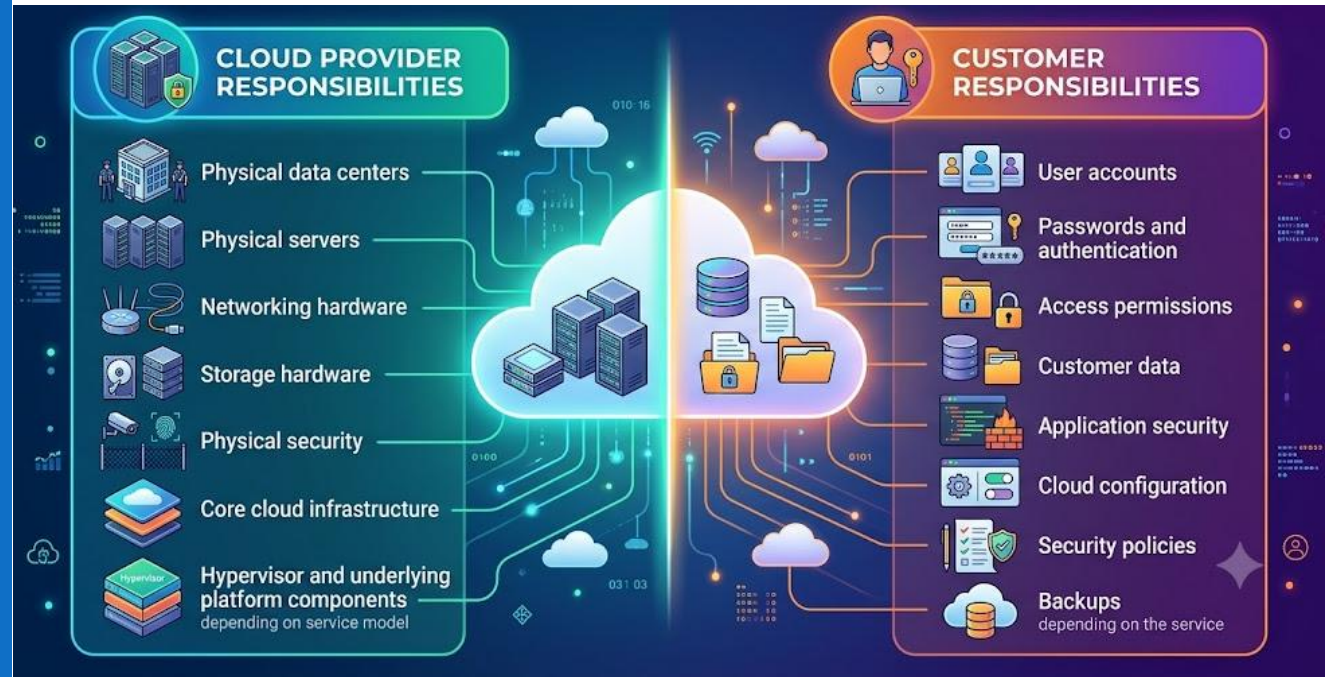
- Monitor user activity
- Monitor network traffic
- Detect unusual behavior
- Review access logs
- Track configuration changes
- Detect security incidents

Auditing

- Audit logs can answer:

Who? → Accessed what? → When? → From where? → What action was performed?

Shared Responsibility Model



Identity and Access Management (IAM)

- **IAM** controls the identities that can access cloud resources and the actions they are allowed to perform.
- It acts as the digital gatekeeper that verifies user identities and controls access to cloud resources.

Core Functions of Cloud IAM

- *Authentication*: Verifies who a user or device is using passwords, biometrics, or Multi-Factor Authentication (MFA).
- *Authorization*: Decides what specific data or services an authenticated user can access via Role-Based Access Control (RBAC).
- *Least Privilege*: Restricts permissions so users and machines only get access to what they need to do their jobs.
- *Auditing & Logging*: Tracks user activity and access events to support compliance and detect suspicious behavior.

Least Privilege Principle

- **Least privilege** means giving an user or application only the permissions required to perform its task.
- It benefits
 - Reduces accidental damage
 - Limits the impact of compromised accounts
 - Prevents unauthorized actions
 - Improves overall security

Disaster Recovery and Backup

Security also includes protecting against data loss and service failures.

- Backup
Create copies of important data.
- Disaster Recovery
Restore applications and services after a major failure.
- Major two objectives:
 - RPO (Recovery Point Objective): How much data loss is acceptable?
 - RTO (Recovery Time Objective): How quickly must the service be restored?

Example:

RPO = 1 hour → At most 1 hour of data loss is targeted.

RTO = 2 hours → Service should be restored within 2 hours.

Cloud Security Architecture

- Cloud security architecture is the strategic framework of policies, technologies, and controls designed to protect systems, data, and applications in the cloud.

